Planet-Typing Near the Radius Valley: Can Sub-Neptunes Masquerade as Terrestrial Worlds in Reflected-Light Spectroscopy?

Abstract: The Habitable Worlds Observatory (HWO) aims to directly image and spectrally characterize Earth-sized planets. A critical challenge for the mission is planet typing: distinguishing truly terrestrial targets from larger, volatile-rich planets that could appear "Earth-like" in low-resolution reflected-light spectra. Transit surveys reveal a statistically significant deficit in planet occurrence near 1.5–2.0 $R_\oplus$ (the "radius valley/Fulton gap"), separating smaller, predominantly rocky planets from larger sub-Neptunes whose radii are inflated by gaseous envelopes (Fulton et al. 2017; Rogers 2015; Lopez & Fortney 2014). In direct imaging, planet radius is not measured from transits, and the observables (planet–star flux ratio and spectral shape) are degenerate with phase, clouds, and atmospheric mean molecular weight, creating a pathway for "un-Earths" to occupy similar observable space as terrestrial planets.

This project will use the open-source radiative transfer framework PICASO to generate a forward-model grid of sub-Neptune reflected-light spectra across FGK host stars (Batalha et al. 2019). The grid will span separations relevant to the Habitable Zone (HZ), heavy-element enrichment (metallicity up to ~$10^3\times$ solar), and cloud-top pressures (Lopez & Fortney 2014). The sub-Neptune grid will be compared against (i) disk-integrated Earth benchmark spectra validated against spacecraft observations (Robinson et al. 2011) and (ii) abiotic terrestrial-planet climate-to-spectra analogs with self-consistent patchy clouds (Windsor et al. 2023). We will identify spectral diagnostics and wavelength-coverage requirements that reduce confusion between gas-rich sub-Neptunes and terrestrial planets, supporting efficient target triage for future direct-imaging surveys.

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1. Introduction:

Kepler-era demographics revealed that small exoplanets are not a smooth continuum: the California-Kepler Survey measured a ≥2× occurrence deficit at 1.5–2.0 $R_\oplus$ for close-in planets, splitting the population into smaller super-Earths and larger sub-Neptunes (Fulton et al. 2017). Mass–radius constraints further indicate that most planets larger than ~1.6 $R_\oplus$ are not purely rocky (Rogers 2015), while thermal-evolution models show that even percent-level H/He envelopes can substantially inflate radii, yielding composition ambiguity near the valley (Lopez & Fortney 2014).

Future direct-imaging missions such as the HWO will observe planets in reflected light and measure their planet-star flux ratios and spectra rather than transit radii. With limited spectral coverage and finite SNR, cloud decks and metallicity/mean-molecular-weight effects can flatten absorption features and reduce scale heights, increasing the likelihood that an envelope-bearing planet is misclassified as terrestrial. This motivates a forward-model grid study that explicitly maps where in parameter space sub-Neptunes become spectrally confusable with terrestrial analogs and identifies which wavelength coverage is most diagnostic.

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2. Objectives and Aims:

Central objective: Determine whether sub-Neptunes near the radius valley can be reliably distinguished from terrestrial planets using reflected-light spectroscopy under HWO-relevant conditions.
Specific aims:

1. Build a forward model grid of sub-Neptune reflected-light spectra varying:
    - Host star type (FGK)
    - HZ-relevant separations/isolation regimes
    - Heavy-element enrichment (metallicity≈1–$10^3\times$ solar)
    - Cloud-top pressure and cloud optical depth
2. Assemble comparison spectra for:
    - Disk-integrated modern Earth benchmarks (Robinson et al. 2011)

    - Abiotic terrestrial secondary-atmosphere analogs with self-consistent clouds (Windsor et al. 2023)
3. Identify wavelength regions/feature combinations that best separate envelope-bearing planets from terrestrial analogs despite cloud and composition degeneracies.
4. Translate results into wavelength-coverage guidance for planet-typing in future direct-imaging surveys.

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3. Methodology

3.1 Sub-Neptune forward modeling with PICASO:

We will use PICASO, an open-source reflected-light radiative transfer model capable of arbitrary phase geometry (Batalha et al. 2019). The model grid will span:

- FGK hosts: representative stellar spectra appropriate for nearby direct-imaging target lists.
- Separations/isolation: inner-to-outer HZ-relevant isolation bins, computed consistently per stellar type.
- Metallicity: up to ~$10^3\times$ solar, motivated by sub-Neptune composition diversity and by the strong dependence of atmospheric structure/spectra on envelope composition (Lopez & Fortney 2014).
- Cloud structure: cloud-top pressure and optical depth as primary degeneracy levers, since clouds can mute features and alter spectral slopes.
- Wavelength range: compute spectra over the full reflected-light range typically used in PICASO applications (e.g., ~0.3–2.5 μm) to capture Rayleigh slopes, and key molecular bands; diagnostics will also be evaluated on mission-relevant sub-ranges.

3.2 Terrestrial comparison set (modern Earth + abiotic analogs):

- Modern Earth benchmark: Use disk-integrated Earth spectra validated against EPOXI spacecraft observations of Earth as an unresolved exoplanet analog, along with a forward model that includes Rayleigh scattering, gas absorption, and cloud effects (Robinson et al. 2011).
- Abiotic terrestrial analogs: Use climate-to-spectra rocky-planet models representing secondary (N_2-dominated) atmospheres across plausible conditions, including self-consistent patchy clouds (Windsor et al. 2023).

3.3 Confusion metrics and planet-typing diagnostics:

We will quantify "confusion" using a combination of physically interpretable diagnostics and statistical similarity under assumed measurement noise floors:

- Band-depth contrasts for major absorbers (e.g., H_2O, CH_4, CO_2)
- Spectral slopes (Rayleigh regime vs cloud-flattened continua)
- Continuum shape/CIA signatures consistent with H_2-rich envelopes
- Similarity metrics define observational degeneracy as spectra that are statistically indistinguishable given realistic uncertainties.

The output will be a mapped "confusion region" in (enrichment, cloud-top pressure, separation, host-star type) space, along with a set of recommended wavelength windows that best reduce ambiguity.

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4. Interdisciplinary Relevance:

- Exoplanet demographics & evolution: the radius valley identifies a population boundary shaped by formation and atmospheric loss.
- Atmospheric physics & radiative transfer: reflected-light spectra encode scattering, molecular absorption, and clouds; PICASO is designed for this regime.
- Astrobiology & life-detection strategy: correct planet typing reduces habitability false positives and improves confidence in interpreting spectra as evidence for (or against) habitable conditions.

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5. Expected Outcomes (Deliverables)**:**

1. A documented library/grid of sub-Neptune reflected-light spectra across FGK hosts, HZ-relevant separations, enrichment, and cloud parameters.

2. Quantitative identification of the most diagnostic spectral regions that discriminate sub-Neptunes from terrestrial/abiotic analogs, including cases where confusion is unavoidable.
3. Wavelength-coverage guidance for planet-typing in future direct-imaging surveys such as HWO.

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6. Timeline:

- Summer 2026 (Months 1–3): Finalize sub-Neptune parameter grid (enrichment, cloud regimes, FGK host bins); implement and validate PICASO reflected-light pipeline; generate primary spectral grid and assemble Earth + abiotic comparison set; compute diagnostic metrics and confusion maps identifying robust spectral discriminants and key wavelength windows.
- Fall 2026 (Month 4): Synthesize results; prepare figures and a short manuscript suitable for conference presentation and external submission (e.g., AAS/DPS).

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7. Team and Expertise:

- Daniel Huang (Undergraduate PI, Ty Robinson's HabLab): Will build and run the PICASO reflected-light forward-model grid, generate diagnostics/confusion maps, and translate results into practical wavelength-coverage recommendations. Daniel has prior experience using PICASO for exoplanet forward modeling and spectral analysis.
- Zarah Brown (Project Mentor, Postdoc at LPL; Ty Robinson's HabLab): Will guide sub-Neptune parameter choices (e.g., enrichment and cloud regimes) and help interpret results in the context of direct-imaging planet-typing.

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8. Budget and Justification: $7,467 total requested.

- $5,967 – Undergraduate summer salary + ERE (13 weeks, 30 hrs/week, $15/hr + 2% Employee Related Expenses). Supports model development, spectral grid generation, analysis, and manuscript preparation.
- $1,500 – Conference travel (AAS/DPS): Airfare ($400), lodging for 4 nights ($700), meals/incidental expenses ($400). Supports the dissemination of results to the planetary science and astrobiology community.

References:

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Windsor, James D., et al. "A Radiative-Convective Model for Terrestrial Planets with Self-Consistent Patchy Clouds." *The Planetary Science Journal*, vol. 4, no. 5, May 2023, p. 94. *DOI.org (Crossref)*, https://doi.org/10.3847/PSJ/acbf2d.